



Principles of Finance

Chapter 11

End-of-Chapter Questions

Stocks and Stock Valuation

Multiple Choice Questions

1. The price-to-earnings (P/E) ratio is _____.
- A. used to calculate a company's book value
 - B. a multiplier used in enterprise valuation
 - C. only applied when valuing preferred stock
 - D. a metric for showing the expectations of the market

Solution: D

Section 11.1 LO 1 Easy

2. The price-to-book (P/B) ratio is also called the _____ ratio.
- A. market-to-book
 - B. enterprise-to-book
 - C. asset-to-price
 - D. liability-to-book

Solution: A

Section 11.1 LO 1 Easy

3. The two primary types of stock valuation multiples are _____.
- A. enterprise value multiples and comparable company value multiples
 - B. equity multiples and enterprise value multiples
 - C. asset value multiples and liability value multiples
 - D. None of the above

Solution: B

Section 11.1 LO 5 Moderate

4. Dividend yield is a form of equity multiple that is primarily used when _____.
- A. the subject company is operating at a loss
 - B. conducting comparisons between companies in different industries
 - C. conducting comparisons between cash returns and investment types
 - D. analyzing mature companies that have been paying dividends for several years

Solution: C

Section 11.1 LO 5 Moderate

5. Dividend yield is computed as the proportion of dividend per share to _____.
A. share price
B. earnings per share
C. market value per share
D. book value per share

Solution: A

Section 11.1 LO 5 Moderate

6. Which of the following statements about enterprise value metrics is NOT true?
A. EV to revenue can be used to assess companies with negative cash flows or firms that are currently experiencing financial losses.
B. ROCE is a profitability ratio that measures the return on the equity in a company in comparison to its financing over a short period of time.
C. EBIT allows investors to assess the core operations of the business without worrying about the costs of the capital structure.
D. EBITDAR is a metric that is used in businesses that have substantial rental and least expenses.

Solution: B

Section 11.1 LO 5 Moderate

7. If an investor's required rate of return increases and all other characteristics of a stock remain the same, the value of the stock will _____.
A. remain the same
B. increase
C. decrease
D. None of the above

Solution: C

Section 11.2 LO 4 Easy

8. A major limitation of the dividend discount model (DDM) is that _____.
A. it is extremely complicated to use
B. it cannot be used with companies that do not pay dividends
C. it must always be used in conjunction with another metric
D. None of the above

Solution: B

Section 11.2 LO 4 Moderate

9. In which of the following ways does preferred stock differ from common stock?

- A. Preferred stock carries voting rights for its ownership.
- B. Preferred stock must be purchased through a broker dealer.
- C. Preferred stock may have a cumulative dividend feature.
- D. In the event of corporate liquidation, preferred stockholders are paid last.

Solution: C

Section 11.4 LO 3 Easy

10. The term *efficient markets* refers to the idea that _____.

- A. publicly traded companies always file their financial reports on time
- B. investors are able to identify underpriced stocks for purchase
- C. stocks trade with minimal costs and prices are current and fair to all traders
- D. financial information on companies and their stock is only available to efficient traders

Solution: C

Section 11.5 LO 1 Easy

Review Questions

1. Briefly discuss one of the primary benefits of using comparative P/E ratios.

Solution: One of the most important benefits of using comparative P/E ratios is that they can standardize stocks with different prices and various earnings levels.

Section 11.1 LO 2 Moderate

2. Name an important characteristic of companies for which the price-to-book (P/B) ratio does not work well.

Solution: The P/B ratio does not usually work very well for companies that primarily consist of intangible assets such as patents, trademarks, and copyrights.

Section 11.1 LO 4 Moderate

3. Briefly describe the main type of scenario in which the two-stage DDM approach might be used to value a firm and its stock.

Solution: The two-stage DDM is designed to value the equity of firms experiencing two stages of growth, an initial period of higher growth and a subsequent period of lower but more stable growth. It is best suited for firms paying residual cash in dividends while having moderate growth.

Section 11.2 LO 4 Moderate

4. Briefly describe a major shortcoming of the zero growth DDM model.

Solution: A major shortcoming is that the model assumes a constant dividend. Because most companies grow over time, dividend payments to shareholders are likely, to increase which does not fit within the model.

Section 11.2 LO 3 Moderate

5. Briefly describe the required inputs for the discounted cash flow (DCF) model.

Solution: The DCF model requires future annual cash flow amounts for each year of the time period being analyzed, as well as an estimated future terminal value that will result from the sale of the stock or asset. A discount rate is also required in order to discount the future cash flows back to the present time.

Section 11.3 LO 2 Moderate

6. Briefly describe preferred stock and some of its ownership advantages compared to common stock.

Solution: Preferred stock is an ownership class of stock that has preferential claims. It generally features a dividend that the company must pay out before dividends are paid to common stockholders, and in case of company bankruptcy and liquidation, owners of preferred shares have priority claim to assets before common stockholders.

Section 11.4 LO 3 Moderate

7. Briefly explain what is meant by the terms *cumulative* and *noncumulative* as they relate to preferred stocks.

Solution: Some preferred stocks are cumulative in dividends, meaning that if a company skips a cash dividend payment, it must pay it at some point in the future. Skipped dividends become a company liability. Other preferred stocks are noncumulative, in which case if the company skips dividends, they are forever lost to the shareholder.

Section 11.4 LO 3 Moderate

8. What were SuperDOT and SOES, and what were they designed to do?

Solution: Both SuperDOT and SOES were computerized trading systems that facilitated the trading of stock in their respective exchanges. SuperDOT was used on the NYSE, while SOES was used on NASDAQ. Both of these systems helped in executing millions of trades every day and helped buyers and sellers obtain the best available prices. On the NYSE, SuperDOT has now been replaced by the Universal Trading Platform, while on NASDAQ, the SOES system has been largely phased out with the emergence of all-electronic trading that increased transaction speed at ever higher trading volumes.

Section 11.5 LO 3 Difficult

9. What are operational efficiency and informational efficiency, and how do they differ in terms of trading markets?

Solution: Operational efficiency concerns the speed and accuracy of processing a buy or sell order at the best available price. Through the years, the competitive nature of the market has promoted operational efficiency. Informational efficiency refers to how quickly a source reflects comprehensive information in the available trading prices. A price is efficient if the market has used all available information to set it, which implies that stocks always trade at their fair value.

Section 11.5 LO 1 Difficult

10. What is meant by informational efficiency, and how does it affect the price of a stock?

Solution: A stock price is considered informationally efficient if the market has used all available information to set it, which implies that stocks always trade at their fair value. If an investor does not receive the most current information, the prices are “stale”; therefore, they are at a trading disadvantage.

Section 11.5 LO 3 Moderate

Problems

1. Today, Sysco Enterprises paid dividends on its common stock of \$1.25 per share. If dividends per share are expected to increase to \$3.50 per share six years from now, what is the percentage dividend growth rate?

Solution: 18.72%

With a financial calculator:

PV = -1.25

N = 6

FV = 3.50

PMT = 0

CPT I/Y = 18.72%

Section 11.2 LO 1 Moderate

2. Let's say you want to purchase shares of Fontaine Ltd. and then hold this stock for six years. The company has a stated dividend policy of \$2.00 annually per share for the next six years, at the end of which time you will sell the stock. You expect to be able to sell the stock for \$35.00 at that time. If you want to earn an 8% return on this investment, what price should you pay today for this stock?

Solution: \$31.30

With a financial calculator:

PMT = 2

N = 6

FV = 35.00

I/Y = 8%

CPT PV = 31.30

Section 11.2 LO 1 Moderate

3. Damian Painting Systems has established a dividend policy of \$3.00 per share per year. If the company plans to be in business forever, what is the value of this stock if an investor wants a 10% return?

Solution: \$30.00

$$\frac{\$3.00}{0.10} = \$30.00$$

Section 11.2 LO 1 Difficult

4. Wilk Productions wants its shareholders to earn a 12% return on their investment in the company. At what value

would Wilk stock be priced if the company paid \$2.75 per share in constant annual dividends forever?

Solution: \$22.92

$$\frac{\$2.75}{0.12} = \$22.92$$

Section 11.2 LO 1 Difficult

5. Dax Industrial Systems has stock currently priced at \$50.00 per share. If investors are earning a 7% return on Dax Industrial, what is the company's annual dividend payment per share?

Solution: \$3.50

$$\frac{\text{Div}}{0.07} = \$50.00$$

Annual Dividend = \$3.50 per share

Section 11.2 LO 1 Moderate

6. If a stock is selling at \$400 with a current dividend of \$40 and a potential investor's required rate of return is 15%, what would be the anticipated dividend growth rate?

Solution: 5%

$$\$400 = \frac{\$40}{0.15 - g}$$

Solving for g and simplifying:

$$0.15 - g = \frac{\$40}{\$400} = 0.10$$
$$g = 0.15 - 0.10 = 0.05, \text{ or } 5\%$$

Section 11.2 LO 1 Difficult

7. Mind Max Inc. has a dividend policy that increases annual dividends by 3% each year. If last year's dividend was \$2.00, the company intends to stay in business for 50 years, and an investor wants a 9% return, what would be the price of Mind Max stock?

Solution: \$32.31

Using the constant growth DDM with a finite horizon:

$$\text{Price} = \$2.00 \times \frac{1.03}{0.09 - 0.03} \times \left[1 - \left(\frac{1.03}{1.09} \right)^{50} \right]$$
$$\text{Price} = \frac{\$2.06}{0.06} \times (1 - 0.944954^{50})$$
$$\text{Price} = \$34.33 \times 0.941043 = \$32.30914 = \$32.31$$

Section 11.2 LO 1 Difficult

8. Odon Corp. paid dividends today in the amount of \$1.50 per share. If Odon will pay dividends of \$5.00 10 years from today, what is the annual dividend growth rate over this 10-year period?

Solution: 12.79%

With a financial calculator:

$$PV = -1.50$$

$$FV = 5.00$$

$$N = 10$$

$$PMT = 0$$

$$CPT I/Y = 0.127945$$

Section 11.2 LO 1 Difficult

9. The Kirkson Distributors common stock is currently selling at \$52.00 per share, pays dividends annually at \$2.50 per share, and has an annual dividend growth rate of 2%. What is the required return?

Solution: 6.81%

$$\$52 = \frac{\$2.50}{r - 0.02}$$

$$r - 0.02 = 0.0481$$

$$r = 0.06481, \text{ or } 6.81\%$$

Section 11.2 LO 1 Difficult

10. If a preferred share of stock pays dividends of \$2.50 per year and the required rate of return for the stock is 6%, what is its intrinsic value?

Solution: \$41.67

$$\frac{\$2.50}{0.06} = \$41.67$$

Section 11.4 LO 2 Moderate

Chapter 11

Test Bank

Stocks and Stock Valuation

True False Questions

1. Using a company's P/E ratio to establish its value is an example of relative valuation.

- A. True
- B. False

Solution: A. P/E ratio is one of the many benchmarks used in relative valuation.

Section 11.1 LO 1 Easy

2. Two key elements of efficient markets are full availability of financial information about relevant companies and full use of this information to make investment decisions.

- A. True
- B. False

Solution: A. These two elements are necessary and extremely relevant for an efficient market to provide the ability to make correct decisions and assumptions.

Section 11.5 LO 5 Moderate

Multiple Choice Questions

1. Stocks do not promise any specified cash flows in the form of coupons or a face value payment at some point in the future. Instead, stocks are attractive to many investors for what reason?

- A. Stocks are safer investments than most other instruments, such as bonds.
- B. Investing in stock has tax advantages for the investor.
- C. Returns on stock are consistently higher than other asset classes, such as bonds.
- D. Stock can be "put" back to the issuing company if the company's condition deteriorates.

Solution: C. Although stock returns can turn negative, over the long run, they exceed returns on bonds and most other asset classes.

Section 11.1 LO 1 Moderate

2. When stock is bought and sold among investors, this is known as what?

- A. Initial public offering
- B. Secondary market
- C. Over-the-counter market
- D. Derivatives market

Solution: B. Once shares are issued in what is known as the primary market, all subsequent trading is done in the secondary market.

Section 11.1 LO 1 Easy

3. What is an advantage of using relative valuation to establish company value?

- A. Detailed knowledge of the company's financial condition is needed.
- B. Value can be determined with certainty.
- C. The technique is simple and quick to apply.
- D. Once it is calculated, the value does not change.

Solution: C. Extensive knowledge of the company is not necessary because components of the valuation technique are readily available, which makes the technique simple and quick to apply.

Section 11.1 LO 1 Moderate

4. A company's P/E ratio is often used by analysts to judge what?

- A. Whether the company is under- or overvalued
- B. The company's liquidity
- C. Whether the company is profitable
- D. The following year's stock price

Solution: A. Acting like a barometer of the market's view of the company, a P/E ratio above or below is often interpreted as a measure of investors' enthusiasm for the stock and can signal that the company is over- or undervalued.

Section 11.1 LO 1 Moderate

5. A company's *market capitalization* refers to what?

- A. The value of the company's stock, calculated as share price times outstanding shares
- B. The value of the company's future dividends
- C. The amount an acquirer would pay to acquire the company
- D. The discount value of the company's cash flows

Solution: A. Market cap refers to the value of the company's stock, calculated as share price times outstanding shares.

Section 11.1 LO 1 Moderate

6. Which of the following is a relative valuation technique used in the market to establish a company's value?
- A. Discount cash flow
 - B. Dividend discount model
 - C. Technical analysis
 - D. Price-to-book (P/B) ratio

Solution: D. The P/B ratio is a common benchmark used to calculate a company valuation.

Section 11.1 LO 1 Moderate

7. Investors frequently analyze a company's dividend yield when deciding whether to invest. What does the dividend yield reveal?
- A. The company's profitability
 - B. The return to the investor from dividends if shares are purchased
 - C. Whether the company will increase or decrease dividends
 - D. The return to the shareholder if shares are held to maturity

Solution: B. The calculation of dividend yield shows the ratio's purpose and the return that will be realized from that dividend. If an investor pays \$X for the share and the share pays a dividend of \$Y, the stock pays, or yields, Y/X to the investor.

Section 11.1 LO 1 Moderate

8. A recent valuation formula uses multiples based on enterprise value (EV). How is EV defined?
- A. Share price times outstanding common shares
 - B. Market capitalization minus cash
 - C. Tangible assets plus intangible assets
 - D. Market capitalization plus net debt

Solution: D. EV is the amount that would need to be paid to acquire a company. This consists of the company's market capitalization, or share price times outstanding shares, plus the amount of debt the company holds, offset by the amount held in cash.

Section 11.1 LO 1 Difficult

9. As a group, dividend discount models (DDMs) are different from relative valuation models. These DDM models are referred to as what?
- A. Intrinsic value models
 - B. Market capitalization
 - C. Technical models
 - D. Residual value

Solution: A. DDMs utilize actual measurable cash flows paid by the company. This cash perspective gives the name of the valuation as *intrinsic*, like the company's cash position.

Section 11.2 LO 2 Moderate

10. DDMs are a form of perpetuity and incorporate the perpetuity valuation formula. Why are DDMs perpetuities?

- A. Dividends are mandatory payments.
- B. Dividends can increase or decrease.
- C. Dividends can theoretically be paid forever.
- D. Dividends are discretionary payments.

Solution: C. Although companies are not required to pay dividends, when dividends are paid, they can theoretically go on forever.

Section 11.2 LO 2 Difficult

11. According to DDMs, if a company increases its dividend payout, the value of the stock price _____.

- A. will increase
- B. will decrease
- C. will remain unchanged
- D. cannot be determined

Solution: A. If dividends rise, the discounted value of the increasing dividend stream must also rise.

Section 11.2 LO 2 Moderate

12. To use a variable, or nonconstant, growth valuation model, the zero-growth model needs to be adjusted to determine what?

- A. Discount rate
- B. Enterprise value
- C. Dividends paid
- D. Time period

Solution: C. Variations to the zero-growth model all relate to the amount of dividends paid per period.

Section 11.2 LO 2 Moderate

13. In a DDM, what is the terminal value?

- A. The discounted value of dividends from the end of a finite growth period to infinity
- B. The dividends paid in the last year of the model
- C. The discounted value of all dividends
- D. The nominal value of all dividends paid

Solution: A. After dividends are forecasted over periods of growth or reduction, the discounted dividends paid over the remaining life of the company are the terminal value.

Section 11.2 LO 2 Difficult

14. An argument is made for companies not to pay dividends. Technology companies and even Warren Buffet follow this reasoning. A company can justify **not** paying dividends because _____.

- A. they are not permitted in all states
- B. investors must pay taxes on dividend income
- C. funds can be reinvested more profitably back into the company
- D. maintaining a stable dividend is difficult

Solution: C. Particularly for growth companies, reinvesting funds leads to faster growth and more dynamic companies, which growth investors seek.

Section 11.2 LO 2 Moderate

15. Discount cash flow (DCF) valuation models are broadly similar to DDMs in that both discount a future flow of funds to establish share price. What is the key difference between them?

- A. DCF discounts company asset values.
- B. DCF identifies and discounts overall company cash flows.
- C. DCF discounts future dividends.
- D. DCF assumes that cash flows can change.

Solution: B. DCF is a more comprehensive valuation approach that goes beyond discounting dividends to incorporating all company sources and uses of cash.

Section 11.3 LO 1 Moderate

16. Using the DCF approach yields the value of the company as a whole. How would one refine this to determine the value of a share of stock?

- A. Divide the company value by total assets.
- B. Divide the company value by the number of outstanding shares.
- C. Subtract debt from company value to get net asset value.
- D. Divide company value by rate of return.

Solution: B. Dividing the company value by the number of outstanding shares gives the share price, or the company's value per share.

Section 11.3 LO 3 Easy

17. What is a limitation of the DCF valuation approach?

- A. Predicting future cash flows from income statement accounts and asset/liability changes is complex.
- B. It cannot be used in multiple industries.
- C. Cash flow is sometimes paid out as dividends.
- D. Cash flows can change materially year to year.

Solution: A. The DCF approach assumes all cash flow, generated from all available sources, that need to be included in the model.

Section 11.3 LO 3 Moderate

18. Preferred stock is an equity instrument but is sometimes referred to as a hybrid. This is because preferred stock also contains features of what?

- A. Bonds
- B. Derivatives
- C. Equity
- D. Assets

Solution: A. Preferred stock holds a priority claim in bankruptcy, a seniority position for dividends, a preestablished dividend payout, and other advantages that are typically characteristic of bonds.

Section 11.4 LO 4 Easy

19. Valuation of preferred stock most closely resembles the _____ valuation method.

- A. relative
- B. technical analysis
- C. constant growth DDM
- D. future cash flow

Solution: C. With growth equal to zero and dividends at a constant rate, the constant growth model is the appropriate model.

Section 11.4 LO 4 Moderate

20. To calculate the return on preferred stock, divide the _____ by the current share price.

- A. annual dividends
- B. discount rate
- C. stock's par value
- D. dividend growth rate

Solution: A. The ratio of the dividend paid to the amount of the investment (share price) shows the stock return, or yield.

Section 11.4 LO 4 Moderate

21. Many preferred stock issues are cumulative in nature. What does this mean?

- A. The shares can be converted into common shares.
- B. Dividends accumulate and are paid at the maturity of the preferred stock.
- C. Any dividends that are not paid in a particular period are carried over and paid in the next period.
- D. The shares accumulate voting rights based on how long they are held.

Solution: C. Unlike common dividends, preferred dividends must be paid. If they are not paid in the current year, they must still be paid in a subsequent period.

Section 11.4 LO 4 Moderate

22. What is an example of the market's operational efficiency?

- A. The Universal Trading Platform used by the NYSE to match buyers and sellers
- B. Access to investment managers at brokerage firms
- C. Floor traders on the NYSE
- D. Many brokers and dealers

Solution: A. A third-generation system for executing buy/sell orders has added efficiency to the settlement of stock trades.

Section 11.5 LO 5 Moderate

23. An investor who adheres to the efficient markets hypothesis is likely to believe which of the following?

- A. Picking individual stocks will uncover undervalued stock opportunities.
- B. Investment opportunities are available that would generate higher returns than indicated by their beta value.
- C. Investment returns and risk are uncorrelated.
- D. It is pointless to search for undervalued stock.

Solution: D. If the market is efficient, especially in its highest form of efficiency, there should be no undervalued stocks because they would already have been purchased and their share prices adjusted.

Section 11.5 LO 5 Difficult